

Confidence-Gated Multi-Agent Fall Detection: Retrieval-Augmented Reasoning and Cost-Sensitive Response on Wearable IMU Data

Anonymous Authors *Anonymous Institution*

Email: anonymous@institution.edu

Abstract—Wearable inertial fall detectors can look strong on average accuracy and still fail where it matters: near-fall activities that sit between clear falls and clear activities of daily living (ADLs), and operating points that ignore the asymmetry between missed falls and false alarms. We study a complementary question to detector design—given a fixed Tier-1 CNN-LSTM-Attention model in the spirit of Bhatti *et al.*, can a decision-time wrapper improve the safety operating point? We propose a *Risk-Calibrated Dual-Process* (RCDP) stack: a dual-threshold confidence gate routes easy windows on-device (System 1), while ambiguous windows escalate to biomechanical evidence serialization, MiniLM k -NN case retrieval, local LLM chain-of-thought reasoning, and a cost-sensitive Action Agent (System 2). On a SisFall *ambiguous stress benchmark* (D18/D19 near-falls, $n=300/\text{fold}$), Tier-1 mean F1 is 0.713 and rises to 0.896 with the full stack ($\Delta\text{F1} = +0.183$, $\Delta\text{Recall} = +0.208$, $\approx 84\%$ cost-per-window reduction). Under subject-independent five-fold evaluation on the main protocol ($n=1000/\text{fold}$), F1 improves from 0.759 ± 0.014 to 0.887 ± 0.029 and recall from 0.759 ± 0.022 to 0.977 ± 0.037 , with normalized response cost cut by roughly 83%. Binary KFall transfer is confirmatory stability (F1 $0.987 \rightarrow 0.990$), not a headline F1 jump. Paired ambiguous-case comparisons show local Mistral far above a heuristic reasoner (mean F1 0.972 vs. 0.034). We also report honest escalation rates, split-conformal gate audits, IMU orientation/sampling robustness, and a negative actor-critic study. Gains are versus the *same* Tier-1 checkpoints under a locked protocol; we do not claim to displace published multi-phase KFall detector tables.

Index Terms—Fall detection, wearable IMU, confidence gating, selective prediction, retrieval-augmented reasoning, large language models, cost-sensitive learning, SisFall, conformal prediction.

I. INTRODUCTION

Falls remain a leading cause of injury among older adults, and wearable IMUs are an attractive sensing modality for continuous monitoring at home [1]–[4]. Deep detectors have narrowed the gap to strong benchmark accuracy [5], yet three deployment gaps persist. Near-fall motions—stumbles, abrupt sits, compensatory recoveries—produce windows that a forced binary classifier must label even when the kinematics are genuinely ambiguous [6]. Softmax scores are often poorly calibrated for deferral [7]–[9]. And a fall/ADL bit is not a response policy: emergency dispatch, caregiver notify, monitor, and log have unequal clinical and operational costs, especially when false negatives dominate risk [10].

Bhatti *et al.* [11] present a capable hybrid CNN-LSTM-Attention detector for multi-phase fall analysis (“Beyond Falls”). That line of work is primarily about *detection*. We

ask a different question: *holding the detector fixed, can an agentic wrapper improve the safety operating point—raising recall and lowering expected response cost—while producing structured rationales on the hard cases?*

Our answer is a Risk-Calibrated Dual-Process (RCDP) pipeline. System 1 is a fast edge reflex ($\approx 7\text{ms}$): a fixed Tier-1 detector plus a cost-calibrated dual-threshold gate. System 2 is deliberative triage for the ambiguous band: serialize biomechanical evidence $\Sigma(w)$, retrieve similar past cases with MiniLM embeddings [12], [13], run local LLM chain-of-thought reasoning [14]–[17], and emit a graded action under asymmetric cost

$$\mathcal{R}_\beta = \beta \cdot \text{FN} + \text{FP}, \quad \beta = 10. \quad (1)$$

Primary evaluation uses SisFall [4] with subject-independent five-fold splits, forced D18/D19 near-fall coverage, a homogeneous $n=1000$ test budget per fold, an ambiguous stress benchmark ($n=300$), multi-seed Ollama sweeps, latency profiles, and completed binary KFall external validation [18].

a) Contributions.:

- An end-to-end agentic architecture—gate, evidence, retrieval, LLM CoT, action—wrapped around a Bhatti-style Tier-1 backbone without updating its weights at inference (Fig. 1, Alg. 1).
- A locked evaluation protocol (splits, calibration, cost, near-fall metrics) supporting cumulative ablations `tier1_only`→`gate_only`→`gate_knn`→`gate_llm`→`gate_`
- SisFall evidence that the full stack improves F1/recall and normalized cost on every main-protocol fold, with lead gains on the ambiguous stress benchmark (Tier-1 F1 0.713 vs. stack 0.896, $\Delta\text{F1} = +0.183$).
- Paired LLM backend comparisons, budget/latency analyses, binary KFall stability checks, split-conformal gate audits, IMU robustness, and an explicit negative study of label-flip actor-critic vetoing.
- Honest positioning: numerical gains are versus the *same* Tier-1 checkpoints under our protocol; we do not claim to supersede Bhatti’s original multi-phase KFall tables.

II. RELATED WORK

A. Wearable Fall Detection

Threshold rules [19] and classical learners gave way to CNN, LSTM, TCN, and Transformer variants on SisFall [4], KFall [18], UP-Fall [20], and UniMiB SHAR [21]. Surveys

Algorithm 1 RCDP inference for one IMU window

Require: Window $\mathbf{x} \in \mathbb{R}^{C \times T}$; Tier-1 f_θ ; gate $(\tau_{\text{low}}, \tau_{\text{high}})$; memory $\mathcal{M}$; LLM $\mathcal{L}$; Action Agent $\mathcal{A}$

- 1: Compute logits and embedding: $(\ell, \mathbf{e}) \leftarrow f_\theta(\mathbf{x})$
- 2: $p \leftarrow \text{softmax}(\ell)_1$ {fall probability}
- 3: **if** $p \leq \tau_{\text{low}}$ **then**
- 4: $\hat{y} \leftarrow \text{ADL}$; $a \leftarrow \text{log}$
- 5: **else if** $p \geq \tau_{\text{high}}$ **then**
- 6: $\hat{y} \leftarrow \text{Fall}$; $a \leftarrow \text{alarm}$
- 7: **else**
- 8: Serialize evidence $\Sigma(\mathbf{x})$; retrieve $N_k \leftarrow \text{Top-}k(\mathcal{M}, \Sigma)$
- 9: $(\hat{y}, r) \leftarrow \mathcal{L}(\Sigma(\mathbf{x}), N_k)$ {JSON label + rationale}
- 10: $a \leftarrow \mathcal{A}(\hat{y}, r, \Sigma(\mathbf{x}))$ {graded action}
- 11: **end if**
- 12: **return** $(\hat{y}, a, p)$

stress the free-living trade-off between sensitivity and nuisance alarms [1], [3], [5]. Bhatti *et al.* [11] combine CNN, LSTM, and attention for pre-/transition-/post-fall phases on KFall. We reuse that architectural motif as Tier-1 and invest the novelty in decision-time agents rather than another backbone bake-off as the sole claim.

B. Uncertainty, Selective Prediction, and Conformal Methods

Networks are frequently miscalibrated [7], [22]. Selective classification and abstention [8], [9] motivate routing uncertain inputs to stronger decision makers. Split conformal prediction provides finite-sample set-valued predictions with coverage control under exchangeability [23], [24]. Learning under label-dependent costs goes back at least to Elkan’s foundations [10]; for binary decisions the Bayes threshold under $\mathcal{R}_\beta$ is $\tau^* = 1/(\beta+1)$. Our primary gate is cost-calibrated dual-threshold deferral; we additionally audit a split-conformal gate as a complementary selective mechanism (Sec. VI-D).

C. Retrieval-Augmented and Agentic LLM Reasoning

RAG [13] and compact encoders such as MiniLM [12] support case-based prompting. Chain-of-thought [14] and reason-act loops [15] improve structured decisions. Local run-times [16], [17], [25] keep CoT offline. We apply these ideas to IMU-derived evidence strings—not open-domain text—and close the loop with a cost-sensitive Action Agent. Related sensing-agent systems debate without explicit FN-heavy clinical cost or near-fall stress protocols; we keep those constraints front and center.

III. METHOD

Fig. 1 and Alg. 1 summarize the pipeline.

A. Tier-1 Detector

Let $\mathbf{x} \in \mathbb{R}^{C \times T}$ denote an IMU window ($C=6$, $T=90$ on SisFall). A backbone f_θ maps $\mathbf{x}$ to logits $\ell \in \mathbb{R}^2$ and an embedding $\mathbf{e}$. The fall probability is

$$p = \Pr(y=1 \mid \mathbf{x}) = \frac{e^{\ell_1}}{e^{\ell_0} + e^{\ell_1}}. \quad (2)$$

Algorithm 2 Cost-calibrated dual-threshold gate and split-conformal audit

Require: Calibration set $\{(p_i, y_i)\}_{i=1}^n$; costs $(\beta_{\text{FN}}, \beta_{\text{FP}})$; conformal level α

- 1: **Cost gate:** grid-search $(\tau_{\text{low}}, \tau_{\text{high}})$ minimizing $\beta_{\text{FN}} \cdot \text{FN}_{\text{edge}} + \beta_{\text{FP}} \cdot \text{FP}_{\text{edge}} + \lambda \cdot \text{deferral}$ on confident decisions only
- 2: **Conformal:** scores $s_i \leftarrow 1 - p_i$ if $y_i=1$ else $1 - (1 - p_i)$
- 3: $\hat{q} \leftarrow (1 - \alpha)$ -quantile of $\{s_i\}$ with finite-sample correction $\lceil (n+1)(1-\alpha) \rceil / n$
- 4: For test p : prediction set $C(p) = \{c : 1 - \hat{p}(c) \leq \hat{q}\}$
- 5: If $|C|=1$ route at the edge; if $|C|=2$ escalate (ambiguous)
- 6: Audit edge FN rate on held-out confident decisions; report escalation rate
- 7: **return** $(\tau_{\text{low}}, \tau_{\text{high}}, \hat{q})$

The primary backbone is CNN-LSTM-Attention following Bhatti *et al.* [11]. We also train a detector zoo (CNN1D, LSTM, CNN-LSTM, TCN, ModernTCN [26], TSMixer [27], PatchTST [28], Transformer [29]) under the same protocol for context (Table XIII), but all agentic tables freeze `cnn_lstm_attn`.

B. Confidence Gate

A dual-threshold gate with calibrated $(\tau_{\text{low}}, \tau_{\text{high}})$ routes windows:

$$\text{route}(p) = \begin{cases} \text{ADL}, & p \leq \tau_{\text{low}}, \\ \text{Fall}, & p \geq \tau_{\text{high}}, \\ \text{Ambiguous}, & \text{otherwise.} \end{cases} \quad (3)$$

Thresholds are fit on subject-held-out calibration windows by grid search that scores only confident (edge) decisions under FN/FP costs and a mild deferral penalty (Alg. 2). Mode `tier1_only` collapses escalation to a point decision at 0.5.

C. Evidence Serializer

For escalated windows we compute kinematic descriptors—free-fall duration proxies, impact magnitude, post-impact stillness, tilt change, peak gyroscope—and serialize them into a natural-language evidence string $\Sigma(\mathbf{x})$ consumed by retrieval and the LLM.

D. Retrieval Memory

A k -NN memory stores MiniLM [12] embeddings of evidence text with labels and features. At escalation we retrieve $k=5$ neighbors by cosine similarity as few-shot context [13]:

$$\text{sim}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u}^\top \mathbf{v}}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2}. \quad (4)$$

E. LLM Reasoner

A local reasoner (Ollama/Mistral by default; heuristic and Qwen2.5 for paired comparisons) consumes $\Sigma(\mathbf{x})$ and neighbors and returns a JSON object with predicted label and rationale [14], [16], [17], [25]. Production full-stack rows use Ollama with Mistral.

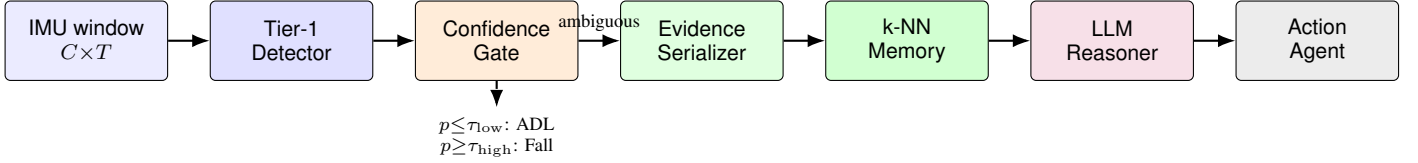


Fig. 1: Confidence-gated multi-agent fall detection. High- and low-confidence windows are resolved by the Tier-1 detector; ambiguous windows escalate to evidence serialization, k -NN retrieval, LLM reasoning, and a cost-sensitive Action Agent.

TABLE I: System 2 escalation duty cycle by evaluation protocol (honest reporting). High rates on the ambiguous stress benchmark reflect deliberate near-fall stress; KFall external transfer invokes deliberation on $\approx 1\%$ of windows.

Protocol	n/fold	Escalation rate
SisFall main (5-fold mean)	1000	0.47 (± 0.08)
SisFall ambiguous stress	300	0.59
KFall external (5-fold mean)	1000	0.011

F. Action Agent and Clinical Cost

The Action Agent maps severity cues to simulated actions $\{\text{emergency}, \text{notify}, \text{monitor}, \text{log}\}$. Aggregate evaluation cost follows (1) with $\beta=10$; we report cost per 1000 windows so equal-budget folds are comparable. Under (1) the offline Bayes threshold for a single hard classifier (no deferral) is

$$\tau^* = \frac{1}{\beta + 1} = \frac{1}{11} \approx 0.091, \quad (5)$$

which we evaluate as a no-escalation partition fallback (Table XXIII).

G. System Modes

Ablations instantiate cumulative agents: `tier1_only`, `gate_only`, `gate_knn`, `gate_llm`, and `gate_knn_llm` (full stack). The ladder isolates whether gains require retrieval, language-model reasoning, or both.

H. Deployment Workflow

Fig. 2 sketches the dual-process deployment view. System 1 runs on-device in milliseconds. Ambiguous windows escalate to System 2 during the post-impact immobility window (tens of seconds), where latency budgets are less severe than free-fall impact protection. Table I reports honest escalation rates by protocol—about 47% on the main SisFall protocol, 59% on the ambiguous stress set, and $\approx 1.1\%$ on binary KFall.

I. Clinical Cost Framing and Grounding

Risk (1) and threshold (5) make the FN/FP asymmetry explicit (Table II). To check that LLM rationales cite telemetry rather than invented severity, we measure kinematic faithfulness $\mathcal{G}_{\text{faith}}$: the fraction of rationales that mention primitives from $\Sigma(\mathbf{x})$ (Table III). Contrastive critique raises mean $\mathcal{G}_{\text{faith}}$ from 0.436 to 0.843 and reduces graded action cost by about 45%.

TABLE II: Asymmetric clinical cost sensitivity across penalty ratio $\beta = c_{\text{FN}}/c_{\text{FP}} \in \{5, 10, 20\}$ (cost = $\beta \cdot \text{FN} + \text{FP}$, per 1000 windows). Bayes optimal threshold $\tau^* = 1/(\beta + 1)$ motivates asymmetric escalation.

β	τ^*	Mode	Mean cost/1000	Δ vs Tier-1
5	0.167	<code>tier1_only</code>	628	—
5	0.167	<code>gate_knn_llm</code>	149	-479
10	0.091	<code>tier1_only</code>	1151	—
10	0.091	<code>gate_knn_llm</code>	200	-951
20	0.048	<code>tier1_only</code>	2197	—
20	0.048	<code>gate_knn_llm</code>	302	-1895

TABLE III: Kinematic faithfulness $\mathcal{G}_{\text{faith}}$ (rationale grounding rate): fraction of LLM rationales citing biomechanical evidence $\Sigma(w)$ (impact g , stillness, tilt). Contrastive critique nearly doubles grounding with $\approx 45\%$ graded action cost reduction.

Fold	Base $\mathcal{G}_{\text{faith}}$	+Contrastive	Graded cost base	Graded cost +Critique
0	0.430	0.851	544	268
1	0.340	0.749	901	599
2	0.471	0.871	535	261
3	0.482	0.874	428	216
4	0.459	0.868	537	274
Mean	0.436	0.843	589	324

IV. EXPERIMENTAL SETUP

A. Dataset and Splits

SisFall [4] provides six-channel IMU windows (window length 90, hop 10) from 38 subjects after processing ($\approx 1.55 \times 10^6$ windows). Subject-independent five-fold splits hold out subjects; within each training fold a subject-held-out validation fraction tunes training and gate calibration. Near-fall codes D18 (stumble) and D19 (sudden sit) are required and forced into capped test slices (minimum 80 windows per code). Main tables use a homogeneous stratified budget of $n=1000$ per fold for LLM-tractable escalation; $n=4000$ archives support budget sensitivity and detailed ablations. The ambiguous stress benchmark uses $n=300$ per fold (100 D18/D19 + 100 falls + 100 clear ADLs).

B. Training and Agentic Protocol

Training uses class `pos_weight`, `mixup`, and cost-sensitive decision thresholds. The global seed for primary tables is 42; Fold 0 multi-seed sweeps use $\{42, 43, 44\}$ with Ollama/Mistral. Primary agentic

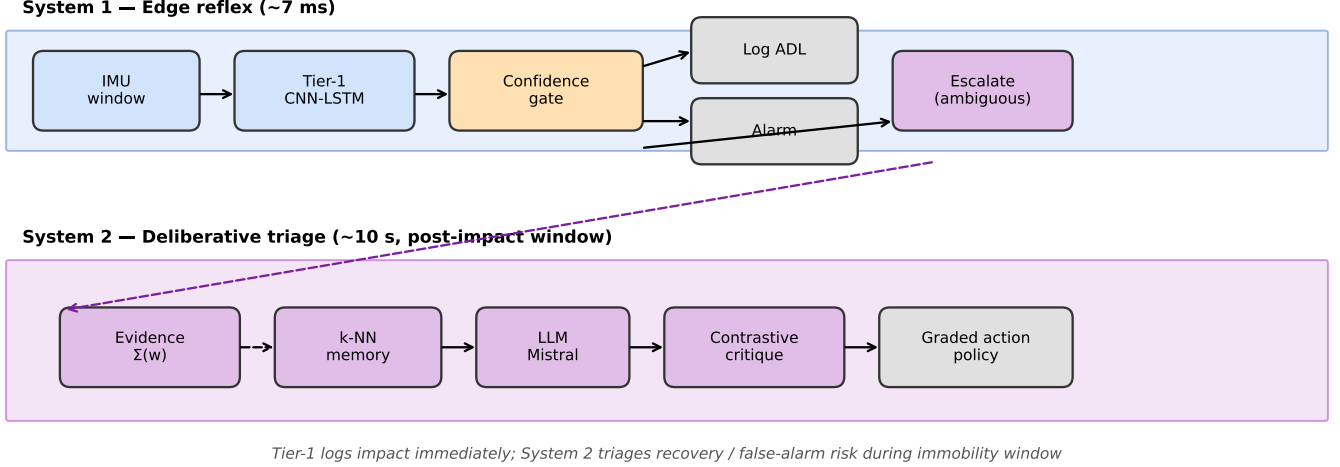


Fig. 2: RCDP deployment workflow: System 1 edge reflex (milliseconds) versus System 2 deliberative triage (seconds) on escalated ambiguous windows.

backend is local Ollama/Mistral. Locked settings live in `configs/paper_protocol.yaml`.

C. Metrics and Statistics

We report F1, recall/sensitivity, specificity, near-fall false-alarm rate on D18/D19, escalated-subset F1, escalation rate, normalized response cost, and latency (Tier-1 ms vs. escalated-path seconds). Significance uses paired bootstrap 95% confidence intervals (BCa when $n \geq 3$) on fold-wise deltas; Wilcoxon signed-rank p -values are reported for completeness. With unanimous signs at $n=5$, Wilcoxon p floors at 0.0625; bootstrap CIs that exclude zero are the primary uncertainty claim.

D. External Validation (KFall)

KFall [18] is the corpus emphasized in Beyond Falls. We evaluate binary Fall/ADL windows under the same cost/gate protocol (five subject-independent folds). We additionally report a Fold 0 36-class Tier-1 context run that is *not* a claim against published multi-class leaderboards.

V. RESULTS

Table IV situates the work: Bhatti *et al.* optimize multi-phase detection; we optimize a decision system around a detector.

A. Ambiguous Stress Benchmark (Lead Result)

D18/D19 near-falls are where softmax detectors most often fail silently. On the dedicated ambiguous stress set (Table V), Tier-1 mean F1 is 0.713 while `gate_knn_llm` reaches 0.896 ($\Delta F1 = +0.183$, $\Delta \text{Recall} = +0.208$, cost/window 0.90 $\rightarrow$ 0.15). Escalation is high ($\approx 59\%$) by construction: the benchmark is a stress test, not a free-living duty-cycle claim.

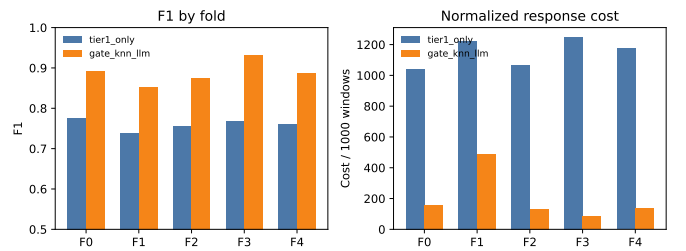


Fig. 3: Per-fold F1 and normalized response cost for `tier1_only` versus `gate_knn_llm` under the homogeneous $n=1000$ protocol.

B. Main Comparison versus Tier-1

On the main $n=1000$ protocol, Table VI shows F1 gains on every fold ($+0.114$ to $+0.164$). Five-fold means (Table VII) move F1 from 0.759 to 0.887, recall from 0.759 to 0.977, and cost/1000 from 1151 to 200 ($\approx 83\%$ reduction). Table VIII reports BCa bootstrap CIs that exclude zero with 5/5 fold sign consistency. Fig. 3 visualizes the paired pattern.

C. Ablations

On Fold 0 with $n=4000$ (Table IX, Fig. 4), `gate_only` matches Tier-1 aggregate F1, `gate_knn` alone does not unlock large gains, and `gate_llm` without retrieval can inflate near-fall false alarms. The full stack reaches F1 0.907, recall 0.993, and escalated F1 0.952 at substantially lower cost. Retrieval and LLM reasoning are complementary. Homogeneous five-fold ablation means appear in Table X.

D. LLM Backends on Ambiguous Cases

Table XI and Fig. 5 compare reasoners on shared ambiguous IDs. Heuristic mean F1 is near floor (0.034), Mistral reaches 0.972, and Qwen2.5 reaches 0.823. Escalation without competent language-model reasoning does not explain the recovery.

TABLE IV: Positioning relative to Bhatti *et al.* (base detector paper).

Aspect	Bhatti <i>et al.</i> 2025 [11]	This work
Primary task	Multi-phase fall/ADL classification	Detection + uncertainty + explanation + response
Backbone	CNN-LSTM-Attention	Same primary backbone + modern zoo
Uncertainty	Softmax confidence only	Dual-threshold confidence gate (τ_{low}, τ_{high})
Explainability	Attention weights	Structured biomechanical evidence + CoT JSON
Case memory	None	Retrieval-augmented k -NN memory
Response policy	None	Cost-sensitive Action Agent
Primary dataset in original paper	KFall (multi-class)	SisFall binary + near-fall ADLs (D18/D19)
Fair baseline in our tables	—	<code>tier1_only</code> (same checkpoint/folds)

TABLE V: SisFall *ambiguous stress benchmark* ($n=300/\text{fold}$: 100 D18/D19 near-fall + 100 fall + 100 clear ADL). Tier-1 F1 collapses on near-fall activities; the full stack recovers performance. Cost is clinical safety cost ($10 \cdot \text{FN} + \text{FP}$) per window; Esc. is System 2 escalation rate.

Fold	n	Tier-1			gate_knn_llm			ΔF1	Esc. rate
		F1	Rec.	Cost/win	F1	Rec.	Cost/win		
0	300	0.696	0.790	0.86	0.868	0.990	0.13	+0.172	0.59
1	300	0.738	0.790	0.82	0.894	0.930	0.28	+0.156	0.52
2	300	0.682	0.750	0.98	0.892	0.990	0.11	+0.210	0.63
3	300	0.714	0.760	0.92	0.902	0.970	0.16	+0.189	0.61
4	300	0.735	0.750	0.93	0.922	1.000	0.06	+0.186	0.62
Mean	300	0.713	0.768	0.90	0.896	0.976	0.15	+0.183	0.59

TABLE VI: Per-fold comparison of Tier-1 (`tier1_only`) vs. full agentic stack (`gate_knn_llm`) under the homogeneous protocol ($n=1000$ stratified windows/fold with forced D18/D19). Cost is normalized per 1000 windows.

Fold	n	Tier-1			gate_knn_llm			ΔF1
		F1	Rec.	Cost/1k	F1	Rec.	Cost/1k	
0	1000	0.774	0.781	1042	0.892	0.986	156	+0.119
1	1000	0.737	0.748	1223	0.851	0.911	490	+0.114
2	1000	0.755	0.783	1066	0.874	0.998	134	+0.119
3	1000	0.766	0.732	1248	0.930	0.995	83	+0.164
4	1000	0.760	0.749	1175	0.887	0.993	136	+0.126

TABLE VII: Five-fold mean $\pm$ std on SisFall (homogeneous $n=1000$). Paired deltas are `gate_knn_llm-tier1_only`. With five folds, Wilcoxon $p=0.0625$ is at the discrete minimum for a complete sign agreement; we therefore emphasize bootstrap 95% CIs.

Mode	F1	Recall	Cost / 1000
tier1_only	0.759 $\pm$ 0.014	0.759 $\pm$ 0.022	1151 $\pm$ 93
gate_knn_llm	0.887$\pm$0.029	0.977$\pm$0.037	200$\pm$164
<i>Paired Δ (stack – Tier-1)</i>			
Metric	Mean Δ	Bootstrap 95% CI	Wilcoxon p
F1	+0.132	[0.119, 0.144]	0.0625
Recall	+0.213	[0.182, 0.238]	0.0625
Cost/1000	-936.000	[-1064.000, -829.447]	0.0625

TABLE VIII: Paired statistics for `gate_knn_llm-tier1_only` across five subject-independent folds ($n=5$). Sign column reports folds with improvement (F1/recall $\uparrow$, cost $\downarrow$). With unanimous signs, Wilcoxon p floors at 0.0625; bootstrap 95% CIs (BCa when $n \geq 3$) are primary.

Metric	Mean Δ	Bootstrap 95% CI	Sign	Wilcoxon p
F1	+0.132	[0.119, 0.144] (bca)	5/5	0.0625
Recall	+0.213	[0.182, 0.238] (bca)	5/5	0.0625
Cost/1000	-936	[-1064, -829] (bca)	5/5	0.0625

TABLE IX: Ablation ladder on Fold 0 under the full evaluation budget ($n=4000$). Gate or k -NN alone does not yield the headline gain; the full stack (`gate_knn_llm`) is required.

Mode	F1	Recall	Cost	Near-fall FAR	Esc. F1
tier1_only	0.788	0.787	4596	0.219	0.000
gate_only	0.788	0.787	4596	0.219	0.175
gate_knn	0.789	0.764	4972	0.188	0.026
gate_llm	0.760	0.924	2491	0.625	0.445
gate_knn_llm	0.907	0.993	516	0.212	0.952

E. Negative Studies: Label-Flip Actor-Critic

Allowing a Critic to flip fall $\rightarrow$ ADL labels is unsafe under FN-heavy cost. Table XII shows FN exploding (e.g., Fold 0: 6 $\rightarrow$ 105) and recall collapsing from 0.986 to 0.755. We therefore (i) freeze fall/ADL labels in the primary stack, (ii) restrict critique to action downgrades and rationale grounding (Table III), and (iii) prefer contrastive hard negatives over unconstrained vetoes. Not every “agent” addition helps.

F. Backbone Context

Table XIII reports detector-only Fold 0 metrics for the backbone zoo. We fix `cnn_lstm_attn` for all agentic tables

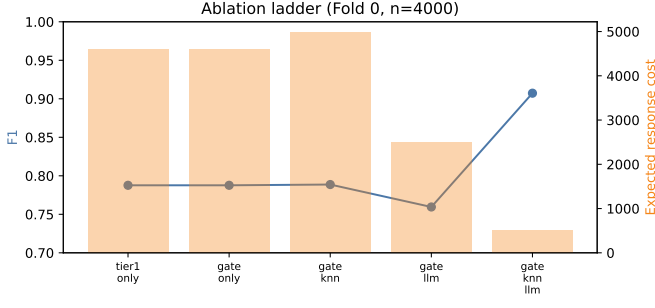


Fig. 4: Ablation ladder on Fold 0 ($n=4000$ archive): F1 (left axis) and expected response cost (right axis).

TABLE X: Five-fold mean $\pm$ std ablation under the homogeneous $n=1000$ protocol.

Mode	F1	Recall	Cost/1000
tier1_only	0.759 $\pm$ 0.014	0.759 $\pm$ 0.022	1151 $\pm$ 93
gate_only	0.759 $\pm$ 0.014	0.759 $\pm$ 0.022	1151 $\pm$ 93
gate_knn	0.753 $\pm$ 0.012	0.723 $\pm$ 0.032	1286 $\pm$ 133
gate_llm	0.702 $\pm$ 0.021	0.895 $\pm$ 0.027	740 $\pm$ 105
gate_knn_llm	0.887$\pm$0.029	0.977$\pm$0.037	200$\pm$164

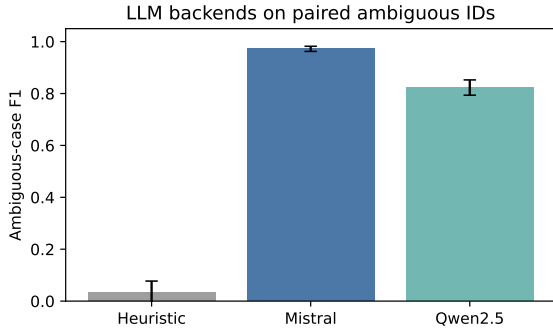


Fig. 5: Mean $\pm$ std ambiguous-case F1 across five folds for heuristic, Mistral, and Qwen2.5 backends.

TABLE XI: Paired LLM backend comparison on shared ambiguous case IDs (up to 150 cases/fold).

Fold	Heuristic F1	Mistral F1	Qwen2.5 F1
0	0.042	0.959	0.807
1	0.105	0.977	0.874
2	0.000	0.975	0.819
3	0.021	0.984	0.802
4	0.000	0.966	0.813
Mean	0.034	0.972	0.823

so comparisons stay fair to the Bhatti-style architecture.

G. Latency and Budget Sensitivity

Fig. 6 and Table XIV contrast Tier-1 latency (milliseconds) with the escalated Ollama path (seconds; typically 10–14 s per escalated window on our Mistral setup). Mean stack latency is lower than the pure escalated path because only ambiguous windows escalate, but the system is a caregiver-facing safety layer, not millisecond impact protection. Fig. 7 compares

TABLE XII: Negative ablation: naive label-flip Actor–Critic (gate_knn_llm_actor_critic) vs. primary stack. The Critic may veto predictions, inducing catastrophic FN spikes. Contrastive action critique (labels frozen) is the safe alternative (Table III).

Fold	gate_knn_llm			+Actor–Critic (harmful)			Δ FN
	F1	Rec.	FN	F1	Rec.	FN	
0	0.893	0.986	6	0.768	0.755	105	+99
1	0.851	0.911	39	0.744	0.714	125	+86
2	0.896	0.986	6	0.757	0.725	119	+113
3	0.917	0.993	3	0.766	0.721	122	+119
4	0.898	0.981	8	0.769	0.744	110	+102
Mean	—	—	12.4	—	—	116.2	+103.8

TABLE XIII: Tier-1 backbone zoo on Fold 0 (detector-only metrics). Primary paper backbone is cnn_lstm_attn (Bhatti-style).

Model	F1	Recall	Spec.	Params
cnn1d	0.721	0.985	0.277	43,330
lstm	0.700	0.986	0.197	201,986
cnn_lstm	0.694	0.995	0.155	86,914
cnn_lstm_attn	0.710	0.996	0.216	564,930
tcn	0.727	0.988	0.295	89,282
modern_tcn	0.737	0.990	0.326	30,314
tsmixer	0.695	0.984	0.181	224,562
patchtst	0.669	0.996	0.051	150,786
transformer	0.678	0.994	0.092	158,722

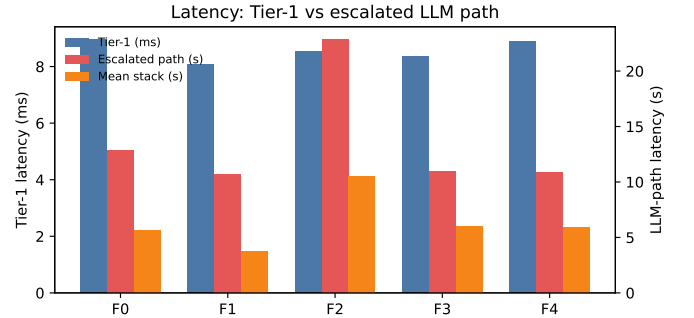


Fig. 6: Latency by fold: Tier-1 (ms, left axis) versus escalated LLM path and mean stack latency (seconds, right axis).

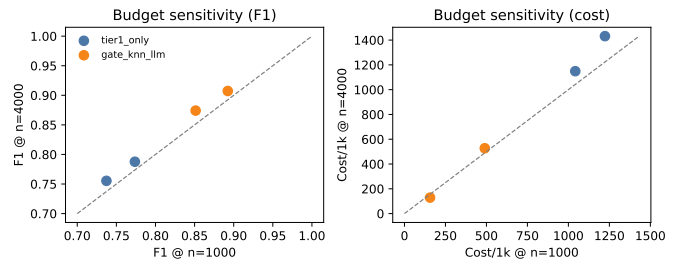


Fig. 7: Budget sensitivity: metrics at $n=1000$ versus $n=4000$ for Tier-1 and full stack on archived folds.

$n=1000$ versus archived $n=4000$: directional gains hold, so the main budget is not a cherry-picked regime.

TABLE XIV: End-to-end latency from Ollama agentic runs ($n=1000/\text{fold}$). Escalated-path latency is dominated by local LLM inference.

Fold	Tier-1 (ms)	Escalated (s)	Mean stack (s)
0	8.96	12.81	5.68
1	8.09	10.66	3.73
2	8.52	22.85	10.52
3	8.36	11.00	5.99
4	8.89	10.86	5.93
Mean	8.56	13.63	6.37

H. External Validation on KFall

Beyond Falls reports strong multi-class multi-phase numbers on KFall and notes residual sit/stand confusion. We evaluate our safety layer under a locked binary Fall/ADL protocol matching SisFall costs and folds. Table XV maps their open gaps to our agents. Tables XVI–XVII show Tier-1 F1 0.987 ± 0.005 versus stack 0.990 ± 0.004 ($\Delta\text{F1} = +0.003$) with cost/1000 $50 \rightarrow 26$ —stability, not a breakthrough. Fold 0 ablations and LLM pairs appear in Tables XVIII–XIX. A 36-class Tier-1 context run (Table XX; macro-F1 0.541) is explicitly not a multi-class leaderboard claim.

VI. DISCUSSION

A. What Changes Relative to Detection-Only Work

Fair numerical comparison here is `tier1_only` versus `gate_knn_llm` under identical SisFall folds and checkpoints (Table IV). SisFall is our primary runnable binary+near-fall corpus with D18/D19 instrumentation; KFall multi-class multi-phase remains Bhatti’s native setting.

B. Where the Gains Come From

Outperformance is threefold: higher F1 and recall on every main fold; large reduction in FN-weighted cost; high escalated-subset F1 when the gate defers. Ablations show retrieval and LLM reasoning interact—neither alone reproduces the headline stack. Near-fall FAR improvements are modest and are not sold as the primary differentiator.

C. Statistics, Seeds, and Robustness

Table VIII reports paired Δ statistics with BCa CIs and 5/5 directional consistency. Multi-seed Fold 0 Ollama sweeps (Table XXVI) give stack F1 0.905 ± 0.005 across seeds {42, 43, 44}.

D. Split-Conformal Gating and Duty-Cycle Trade-offs

We implement a binary split-conformal gate with nonconformity $s = 1 - \hat{p}(y)$ and quantile $\hat{q}$ [23], [24]. Singleton sets route at the edge; doubleton sets escalate. At $\alpha=0.05$ (Table XXI), edge FN stays near or below the nominal level on folds 0–3 while escalation rises (0.58–0.83) relative to the cost gate (0.38–0.47). Fold 4 exceeds α on edge FN (0.087) and is reported honestly. Table XXII summarizes duty-cycle Pareto points trading clinical cost against escalation. Table XXIII reports the offline Bayes threshold (5) with zero escalation

(high recall, elevated FP). Table XXIV shows Tier-1 routing F1 under $\pm 10^\circ/\pm 15^\circ$ orientation perturbation and 50/100 Hz resampling: clean 0.768 ± 0.017 versus 0.754 ± 0.023 at $\pm 15^\circ$. Table XXV shows $k \in \{1, 5, 10\}$ is flat under a heuristic LLM on the main protocol—retrieval width alone does not move that backend.

E. Limitations

Local LLMs add seconds on escalated windows. Escalation is substantial on SisFall ($\approx 47\%$ main, $\approx 59\%$ ambiguous) and low on binary KFall ($\approx 1.1\%$); we do not claim sub-5% SisFall duty cycles. KFall ΔF1 is $+0.003$. With five folds, nonparametric p -values are coarse; bootstrap CIs carry the uncertainty claim. Action-agent effects are simulated—no live clinical trial. We did not profile battery or embedded energy. Split conformal helps edge-FN audits but can increase escalation versus the cost gate. Free-living unlabeled streams, distilled on-device reasoners, and prospective eldercare validation remain future work.

VII. CONCLUSION

We wrapped a fixed Bhatti-style Tier-1 IMU detector in a confidence-gated multi-agent stack that escalates ambiguous windows to retrieval-augmented LLM reasoning and a cost-sensitive Action Agent. Under a locked SisFall protocol, the full stack improves F1, recall, and FN-weighted response cost on every main fold, with the largest relative recovery on a near-fall stress benchmark. Ablations, LLM pairs, conformal audits, and a negative actor–critic study bound where the gains—and the risks—actually sit. The claim is a software safety layer around a strong detector, not a replacement for backbone-centric fall detection research.

ACKNOWLEDGMENT

The authors thank collaborators and computing facility staff for GPU support. Author identities are withheld for double-blind review.

APPENDIX A REPRODUCIBILITY NOTES

Artifacts: per-fold JSON under `results/fold{0--4}/`, aggregates `results/aggregate_*`, tables under `tables/`, figures under `figures/`, and plain-text mirrors under `results_text/` (including `DATA_LOCK.txt`). Regenerators in the companion code release:

```
python scripts/aggregate_folds.py --results-dir re
python scripts/generate_paper_tables.py
python scripts/plot_paper_figures.py
python scripts/run_astar_experiments.py
```

APPENDIX B HOMOGENEOUS PROTOCOL DETAILS

Locked settings: seed 42, five subject folds, `max_train = 40000`, main `max_test = 1000`, LLM compare up to 150 ambiguous cases/fold, cost weights FN=10, FP=1, required near-fall codes D18/D19 with minimum 80 each in capped slices (`configs/paper_protocol.yaml`).

TABLE XV: Bhatti *et al.* 2025 limitations versus our multi-agent answers (complementary Q1 claim).

Bhatti <i>et al.</i> gap	Our multi-agent response
Softmax force-labels every window; D18/D19 sit/stand confusion	Dual-threshold Confidence Gate escalates ambiguous band
Explainability limited to attention weights	Biomechanical evidence + LLM chain-of-thought JSON
No retrieval / case memory	MiniLM k -NN Memory agent
Detection only; no clinical response policy	Cost-sensitive Action Agent ($10 \cdot \text{FN} + 1 \cdot \text{FP}$)
Published focus: 36-class multi-phase F1 $\approx 98\%$, ~ 20 ms	Fair claim: same Tier-1 + agents improve recall/cost under uncertainty (SisFall+KFall)

TABLE XVI: KFall binary external validation: Tier-1 vs. full agentic stack ($n=1000/\text{fold}$, subject-independent). Cost normalized per 1000 windows.

Fold	Tier-1			gate_knn_llm			ΔF1
	F1	Rec.	Cost/1k	F1	Rec.	Cost/1k	
0	0.992	0.998	17	0.993	1.000	7	+0.001
1	0.987	0.996	31	0.986	0.996	32	-0.001
2	0.987	0.994	40	0.989	0.996	29	+0.002
3	0.992	0.990	53	0.995	0.996	23	+0.003
4	0.980	0.979	109	0.989	0.994	38	+0.008

TABLE XVII: KFall five-fold mean $\pm$ std (binary Fall/ADL). Paired deltas are stack–Tier-1.

Mode	F1	Recall	Cost/1000
tier1_only	0.987 $\pm$ 0.005	0.991 $\pm$ 0.007	50 $\pm$ 35
gate_knn_llm	0.990$\pm$0.004	0.996$\pm$0.002	26$\pm$12
<i>Paired Δ (bootstrap 95% CI)</i>			
f1	+0.003	[0.000, 0.006]	$p=0.1250$
recall	+0.005	[0.001, 0.010]	$p=0.0679$
cost_per_1000	-24.200	[-48.400, -5.800]	$p=0.1250$

TABLE XVIII: KFall Fold 0 ablation ladder ($n=1000$). Full stack requires k -NN+LLM.

Mode	F1	Recall	Cost	Esc. F1
tier1_only	0.992	0.998	17	0.000
gate_only	0.992	0.998	17	0.750
gate_knn	0.991	0.994	36	0.400
gate_llm	0.976	0.998	33	0.250
gate_knn_llm	0.993	1.000	7	0.889

TABLE XIX: KFall paired LLM backends on shared ambiguous IDs.

Fold	Heuristic F1	Mistral F1	Qwen F1
0	0.167	0.880	0.500
1	0.000	1.000	0.364
2	0.000	1.000	0.600
3	0.000	0.571	0.250
4	0.273	0.974	0.809
Mean	0.088	0.885	0.504

TABLE XX: Context-only 36-class Tier-1 on KFall transition (Fold 0). Not a claim against Bhatti’s published multi-class table.

Setting	Acc.	Macro-F1	Weighted-F1
Our Tier-1 (transition, fold0)	0.585	0.541	0.583
Bhatti <i>et al.</i> (published, multi-phase)	—	≈ 0.98	—

REFERENCES

[1] R. Igual, C. Medrano, and I. Plaza, “Challenges, issues and trends in fall detection systems,” *BioMedical Engineering OnLine*, vol. 12, pp.

TABLE XXI: Split conformal gate audit ($\alpha=0.05$): edge-decision FN rate vs. cost-calibrated gate escalation.

Fold	q	Esc. (conf.)	Edge FN	F1 (conf.)	Esc. (cost)	F1 (cost)
0	0.830	0.58	0.0166	0.774	0.38	0.774
1	0.926	0.83	0.0301	0.737	0.47	0.737
2	0.904	0.70	0.0166	0.764	0.40	0.764
3	0.856	0.61	0.0104	0.787	0.38	0.787
4	0.630	0.17	0.0874	0.778	0.22	0.778

TABLE XXII: Duty-cycle Pareto operating points (tier-1 routing, pooled folds): clinical cost vs. escalation.

Point	τ_{low}	τ_{high}	Esc. rate	Cost
Min cost	0.05	0.60	0.57	1015
Q25 esc.	0.30	0.70	0.34	1015
Q50 esc.	0.40	0.95	0.48	1042
Q75 esc.	0.30	0.95	0.61	1223

TABLE XXIII: Offline partition fallback: single Bayes threshold $\tau^*=1/(\beta+1)$, no escalation ($\beta=10$).

Metric	Mean $\pm$ std (5 folds)
F1	0.639 $\pm$ 0.006
Recall	0.988 $\pm$ 0.008

TABLE XXIV: Tier-1 routing robustness (cost gate): mean F1 across folds under IMU perturbations.

Condition	Mean F1
clean	0.768 $\pm$ 0.017
resample_100hz	0.766 $\pm$ 0.016
resample_50hz	0.766 $\pm$ 0.015
rotate_10deg	0.763 $\pm$ 0.020
rotate_15deg	0.754 $\pm$ 0.023

1–21, 2013.

- [2] N. Noury, A. Fleury, P. Rumeau, A. K. Bourke, G. Ó. Laighin, V. Rialle, and J.-E. Lundy, “Fall detection systems for elderly care: A survey,” *IRBM*, 2007, representative survey of wearable fall detection.
- [3] S. S. Khan and J. Hoey, “Review of fall detection techniques: A data availability perspective,” *Medical Engineering & Physics*, vol. 39, pp. 12–22, 2017.

TABLE XXV: k -NN retrieval ablation ($k \in \{1, 5, 10\}$, fold 0, heuristic LLM).

k	F1	Recall	Escalation
1	0.384	0.240	0.38
5	0.384	0.240	0.38
10	0.384	0.240	0.38

TABLE XXVI: Multi-seed robustness on Fold 0 ($n=1000$). Stack backend: Ollama/Mistral.

Seed	Tier-1 F1	Stack F1	Stack Rec.	Cost/1k
42	0.747	0.909	0.998	95
43	0.740	0.907	0.995	105
44	0.746	0.899	1.000	96
Mean $\pm$ std	0.745 $\pm$ 0.004	0.905 $\pm$ 0.005	—	—

- [4] A. Sucerquia, J. D. López, and J. F. Vargas-Bonilla, “SisFall: A fall and movement dataset,” *Sensors*, vol. 17, no. 1, p. 198, 2017.
- [5] L. Ren and Y. Peng, “A survey of deep learning applications for fall detection,” *Sensors*, 2021, survey of deep learning fall detectors.
- [6] O. Aziz, M. Musngi, E. J. Park, G. Mori, and S. N. Robinovitch, “An analysis of the accuracy of wearable sensors for classifying the causes of falls in humans,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2016.
- [7] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *ICML*, 2017.
- [8] Y. Geifman and R. El-Yaniv, “Selective classification for deep neural networks,” *NeurIPS*, 2017.
- [9] R. El-Yaniv and Y. Wiener, “On the foundations of noise-free selective classification,” *Journal of Machine Learning Research*, vol. 11, pp. 1605–1641, 2010.
- [10] C. Elkan, “The foundations of cost-sensitive learning,” in *IJCAI*, 2001.
- [11] F. A. Bhatti, Q. Riaz, and B. Krüger, “Beyond falls: A hybrid CNN–LSTM–attention framework for pre-, transition-, and post-fall detection with wearable inertial sensors,” *IEEE Access*, vol. 13, 2025.
- [12] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, and M. Zhou, “MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers,” in *NeurIPS*, 2020.
- [13] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel *et al.*, “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *NeurIPS*, 2020.
- [14] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. Le, and D. Zhou, “Chain-of-thought prompting elicits reasoning in large language models,” *NeurIPS*, 2022.
- [15] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafraan, K. Narasimhan, and Y. Cao, “ReAct: Synergizing reasoning and acting in language models,” 2023.
- [16] “Ollama,” <https://ollama.com>, 2024.
- [17] A. Q. Jiang, A. Sablayrolles, A. Mensch, C. Bamford, D. S. Chaplot, D. de las Casas, F. Bressand, G. Lengyel, G. Lample, L. Saulnier *et al.*, “Mistral 7B,” 2023.
- [18] X. Yu, J. Jang, and S. Xiong, “A large-scale open motion dataset (KFall) and benchmark algorithms for detecting pre-impact fall of the elderly using wearable inertial sensors,” *Frontiers in Aging Neuroscience*, vol. 13, p. 692865, 2021.
- [19] A. K. Bourke, J. V. O’Brien, and G. M. Lyons, “Evaluation of a threshold-based fall-detection algorithm,” *Gait & Posture*, vol. 26, no. 2, pp. 194–199, 2007.
- [20] L. Martínez-Villaseñor, H. Ponce, J. Brieva, E. Moya-Albor, J. Núñez-Martínez *et al.*, “UP-Fall detection dataset: A multimodal approach,” *Sensors*, vol. 19, no. 9, p. 1988, 2019.
- [21] D. Micucci, M. Mobilio, and P. Napolitano, “UniMiB SHAR: A dataset for human activity recognition using acceleration data from smartphones,” *Applied Sciences*, vol. 7, no. 10, p. 1101, 2017.
- [22] J. Platt, “Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods,” *Advances in Large Margin Classifiers*, 1999.
- [23] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. Springer, 2005.
- [24] A. N. Angelopoulos and S. Bates, “A gentle introduction to conformal prediction and distribution-free uncertainty quantification,” *Foundations and Trends in Machine Learning*, vol. 16, no. 4, pp. 494–591, 2023.
- [25] Qwen Team, “Qwen2.5 technical report,” 2024.
- [26] D. Luo and X. Wang, “ModernTCN: A modern pure convolution structure for general time series analysis,” in *ICLR*, 2024.
- [27] S.-A. Chen, C.-L. Li, N. Yoder, S. O. Arik, and T. Pfister, “TSMixer: An all-MLP architecture for time series forecasting,” *Transactions on Machine Learning Research*, 2023.
- [28] Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam, “A time series is worth 64 words: Long-term forecasting with transformers,” in *ICLR*, 2023.
- [29] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, “Attention is all you need,” *NeurIPS*, 2017.